# SigmaFlow — Automated DMAIC Analysis Report

Automated Lean Six Sigma / DMAIC Analysis

**Engine:** SigmaFlow Engine v8

09 March 2026, 19:45

# Abstract

Automated SPC analysis of 'manufacturing_process_dataset' (300 observations, 7 variables) by SigmaFlow v10. 1 statistical anomaly(ies) detected (1 critical, 0 warnings). Regression R?=0.084. 1 column(s) flagged as non-normal.

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# 1 Introduction

This report was automatically generated by **SigmaFlow v8**, a modular Python framework for industrial process analysis. A total of **1 dataset(s)** were processed through the complete analysis pipeline.

The analysis pipeline consists of the following stages:

1. **Dataset Detection** — Automatic classification of dataset type (SPC, capability, logistics, Pareto, service, DOE).
2. **Statistical Analysis** — Descriptive statistics, control limits, capability indices (Cp, Cpk), and trend detection.
3. **Visualization** — Control charts, histograms, Pareto charts, capability plots, and correlation heatmaps.
4. **Root Cause Detection** — Pearson and Spearman correlation analysis to identify variables most associated with defects or variation.
5. **Insight Generation** — Western Electric Rules and statistical rules evaluated against the data; insights exported as JSON.
6. **Report Generation** — This document: fully annotated with interpretations and recommendations.

| Dataset | Type | Shape | Time (s) | Insights |
|---|---|---|---|---|
| manufacturing_process_dataset | SPC | 300×7 | 7.47s | 2 |

# 2 Methodology

The following statistical methods and tools are employed in this analysis:

### Statistical Process Control (SPC)

Control charts are used to distinguish between *common-cause variation* (inherent randomness) and *special-cause variation* (assignable, non-random events). The **XmR chart** (Individuals and Moving Range) is used for individual observations. Control limits are set at $\mu \pm 3\sigma$.

### Western Electric Rules

Four detection rules are applied to identify non-random patterns:

- **Rule 1** — Any single point outside $\pm 3\sigma$ limits (probability $\approx 0.27\%$ under normality).
- **Rule 2** — Nine consecutive points on the same side of the center line (probability $\approx 0.4\%$).
- **Rule 3** — Six consecutive points trending upward or downward (indicates drift).
- **Rule 4** — Fourteen consecutive points alternating direction (indicates over-adjustment or dual-stream process).

### Process Capability Indices

When specification limits are available, the following indices are computed:

| Index | Formula | Interpretation |
|---|---|---|
| $C_p$ | $(USL - LSL)/6\sigma$ | Potential capability (centered) |
| $C_{pk}$ | $\min(C_{pu}, C_{pl})$ | Actual capability (off-centering penalty) |
| $C_{pu}$ | $(USL - \mu)/3\sigma$ | Upper one-sided capability |
| $C_{pl}$ | $(\mu - LSL)/3\sigma$ | Lower one-sided capability |

**Capability thresholds:**

- $C_{pk} < 1.00 \rightarrow$ Process **not capable**
- $1.00 \leq C_{pk} < 1.33 \rightarrow$ **Marginal** capability
- $1.33 \leq C_{pk} < 1.67 \rightarrow$ **Acceptable** capability
- $C_{pk} \geq 1.67 \rightarrow$ **Excellent** (Six Sigma level)

### Root Cause Analysis

Pearson and Spearman rank correlations are computed between all process variables and the primary quality output. Variables are ranked by absolute correlation to identify the most likely process drivers.

| **\|r\| threshold** | **Strength** |
|---|---|
| $\|r\| \geq 0.70$ | Strong association — high priority investigation |
| $0.50 \leq \|r\| < 0.70$ | Moderate association — medium priority |
| $0.30 \leq \|r\| < 0.50$ | Weak association — monitor |
| $\|r\| < 0.30$ | Negligible |

# 3 Dataset Analysis

## 3.1 manufacturing_process_dataset — SPC

### Dataset Overview

| | |
|---|---|
| **Dataset** | **manufacturing_process_dataset** |
| **Type** | **SPC** |
| **Dimensions** | 300 rows × 7 columns |
| **Processing** | 7.47 seconds |

### Summary

Automated SPC analysis of 'manufacturing_process_dataset' (300 observations, 7 variables) by SigmaFlow v10. 1 statistical anomaly(ies) detected (1 critical, 0 warnings). Regression R?=0.084. 1 column(s) flagged as non-normal.

### Statistical Analysis Results

| Metric | Value |
|---|---|
| **n** | 300 |
| **mean** | 79.8672 |
| **std** | 3.0066 |
| **x_chart.UCL** | 88.887 |
| **x_chart.CL** | 79.867 |
| **x_chart.LCL** | 70.848 |
| **x_chart.n_ooc** | 2 |
| **mr_chart.UCL** | 11.26 |
| **mr_chart.CL** | 3.447 |
| **mr_chart.n_ooc** | 2 |
| **trend.direction** | stable |
| **trend.tau** | -0.0175 |
| **trend.p_value** | 0.7629 |

## Process Stability Analysis

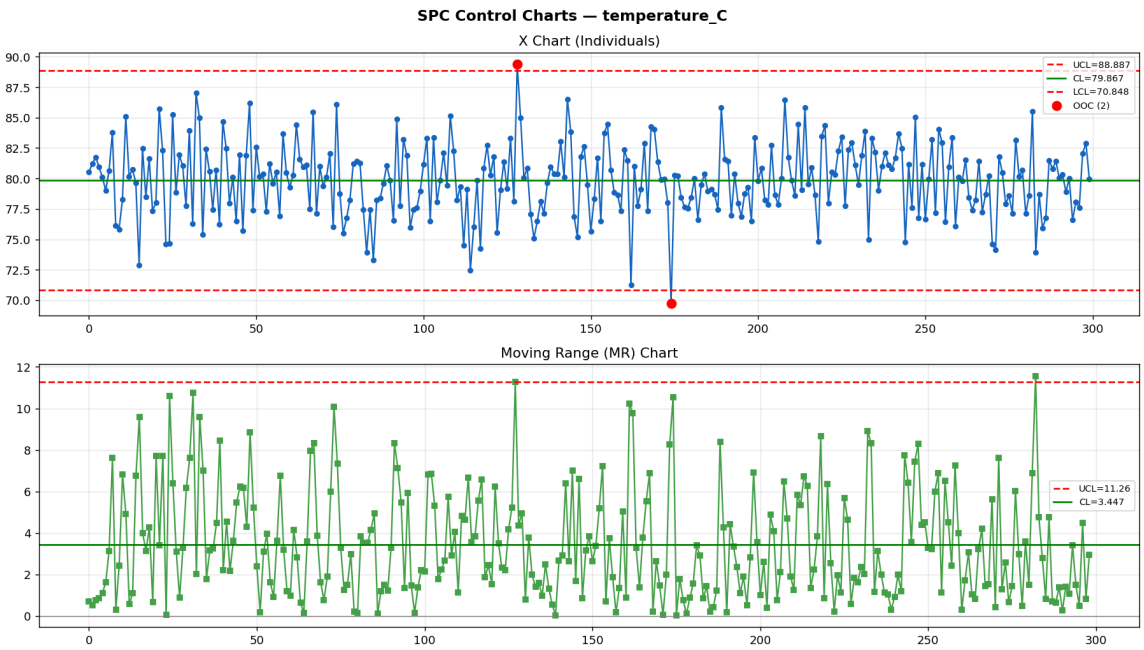


Figure 1: Spc Xmr Chart

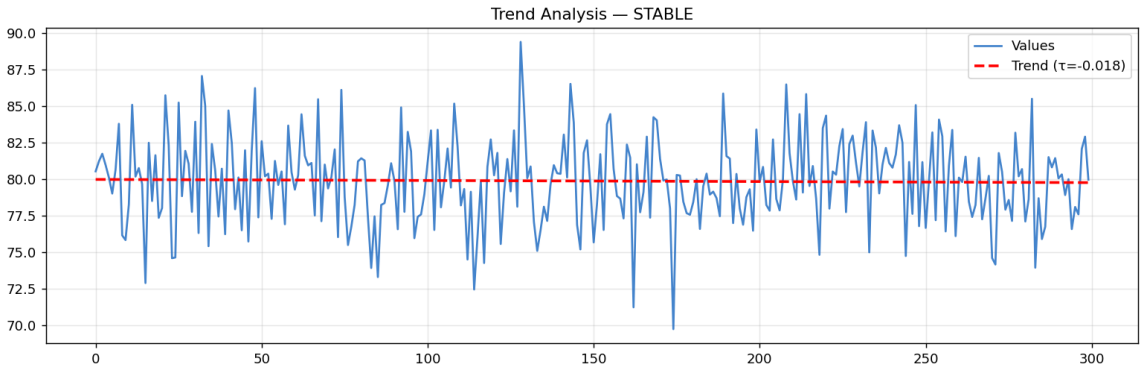


Figure 2: Spc Trend

## Graphical Results

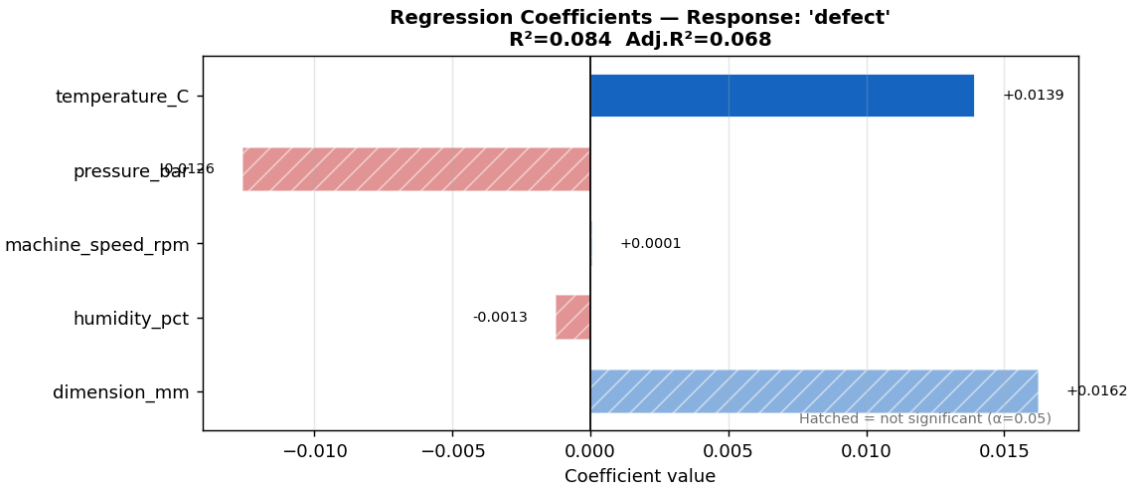


Figure 3: Regression Coefficients

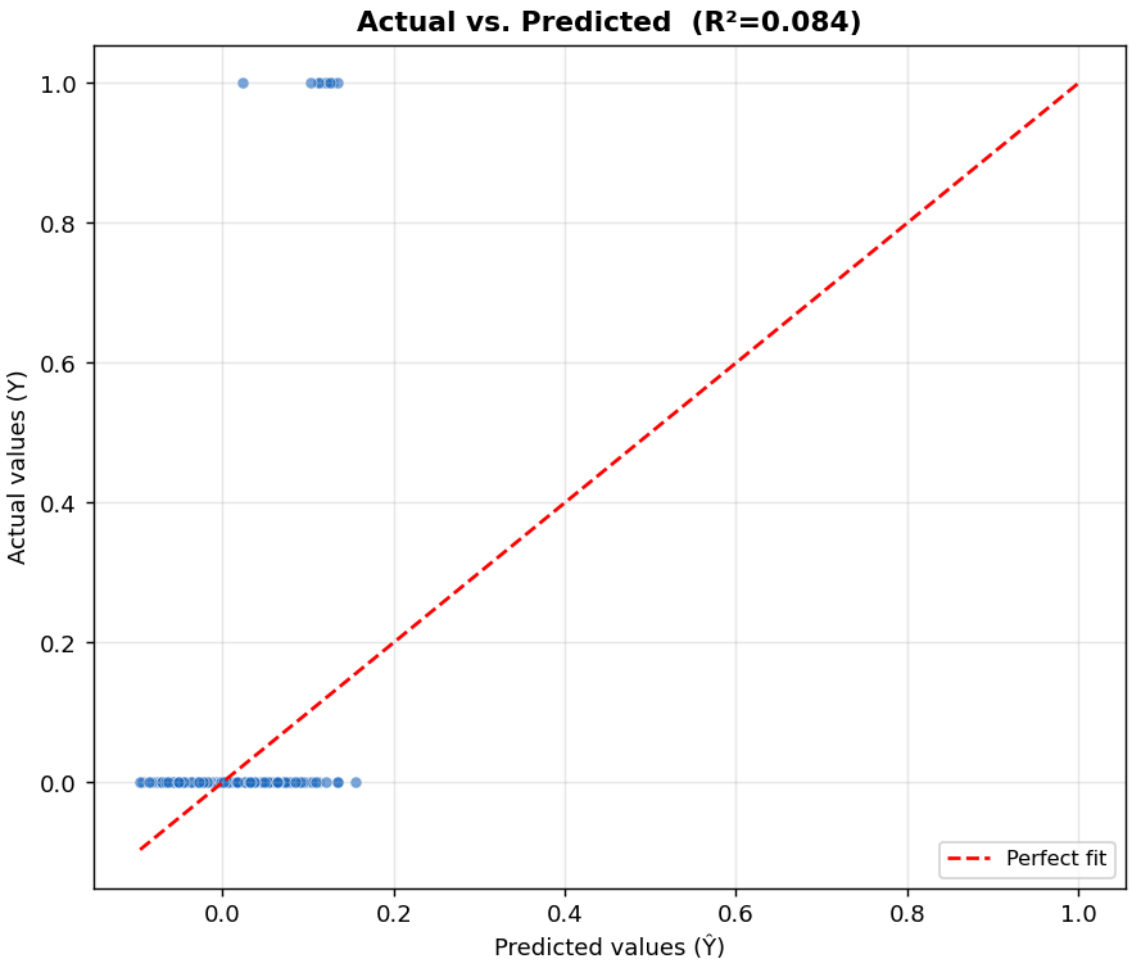


Figure 4: Regression Actual Vs Predicted

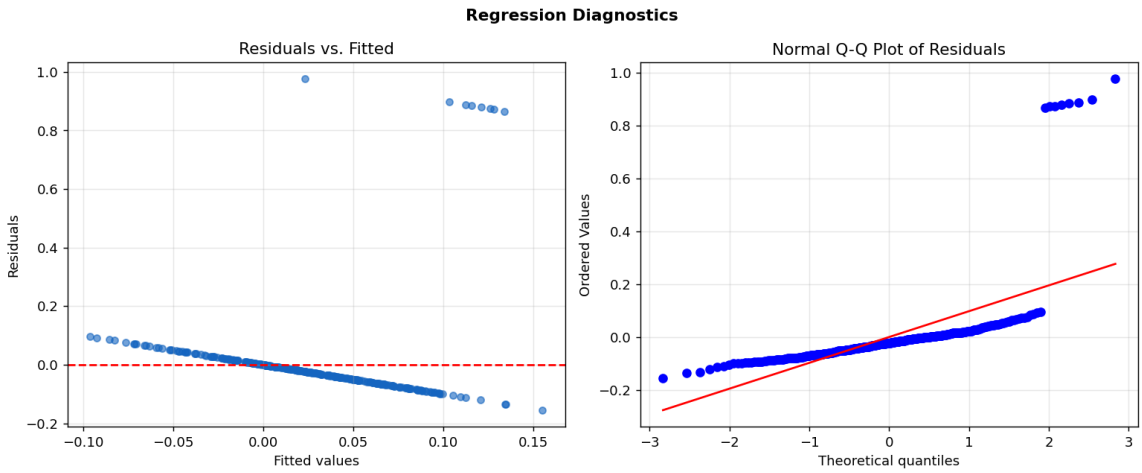


Figure 5: Regression Diagnostics

## Detected Anomalies & Statistical Insights

### Critical Findings Western Electric Rule 1

- **Finding:** Rule 1: 2 point(s) outside the $3\sigma$ control limits (at indices [128, 174]).
- **Interpretation:** A data point beyond $\pm 3\sigma$ from the mean is statistically rare (~0.27% chance under normality). Its presence strongly suggests a special-cause event — something changed in the process at that moment.
- **Recommended Investigation:** Identify and investigate the specific points flagged. Check for equipment malfunction, operator changes, raw material variation, or measurement errors. Remove confirmed special causes before computing capability indices.

### Informational Trend Stable

- **Finding:** No statistically significant trend detected in the process data.
- **Interpretation:** The Spearman correlation coefficient ($\tau$ = -0.018) and p-value (p = 0.7629) indicate no systematic drift in the process over time.
- **Recommended Investigation:** Continue routine monitoring. The process is temporally stable.

## Root Cause Indicators

Correlation analysis was performed using the variable **defect** as the quality target. No variables showed meaningful correlation with 'defect'. The variation may be driven by factors not present in this dataset.

### Variable Importance Ranking

| Variable | Pearson r | Spearman r | Strength |
|---|---|---|---|
| temperature_C | **+0.263** | +0.217 | Negligible |
| pressure_bar | **-0.079** | -0.089 | Negligible |
| machine_speed_rpm | **+0.058** | +0.061 | Negligible |
| dimension_mm | **+0.020** | -0.004 | Negligible |

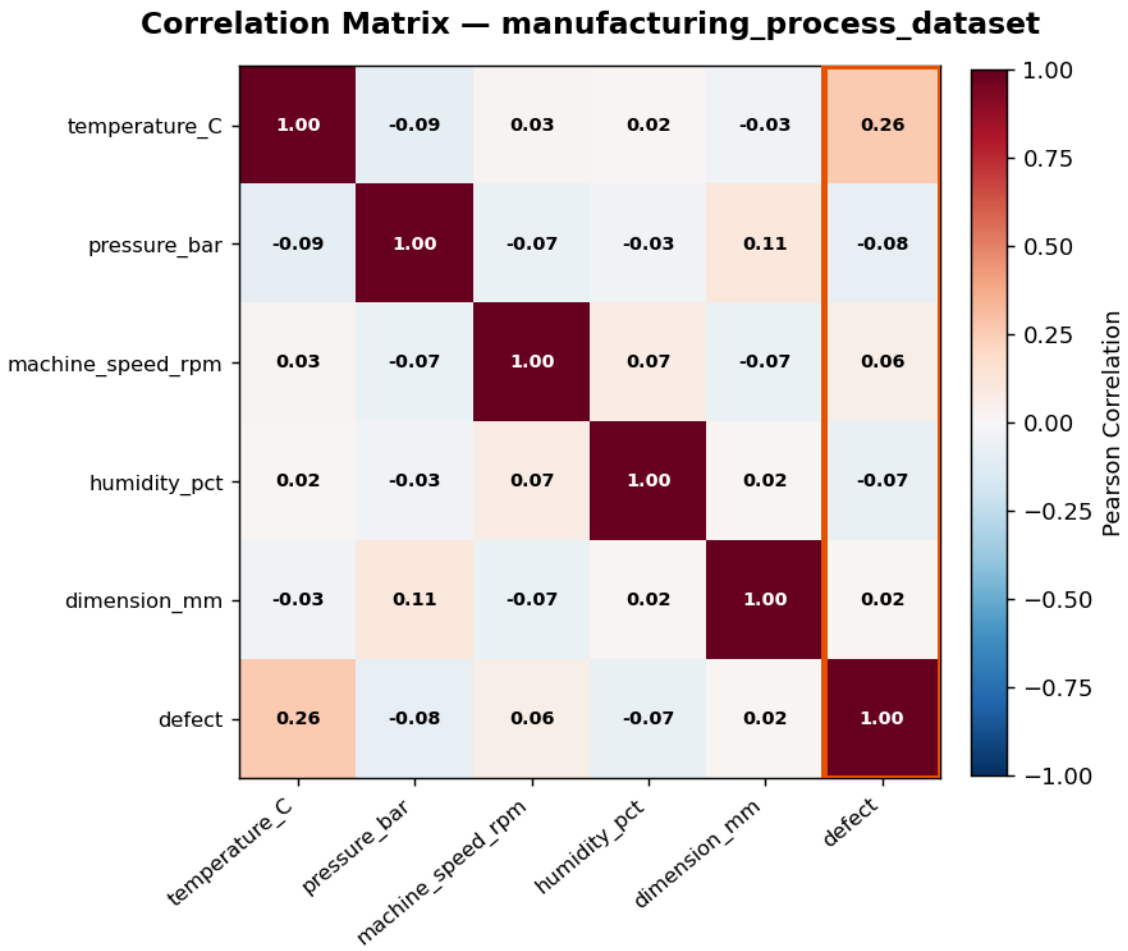


Figure 6: Correlation Heatmap

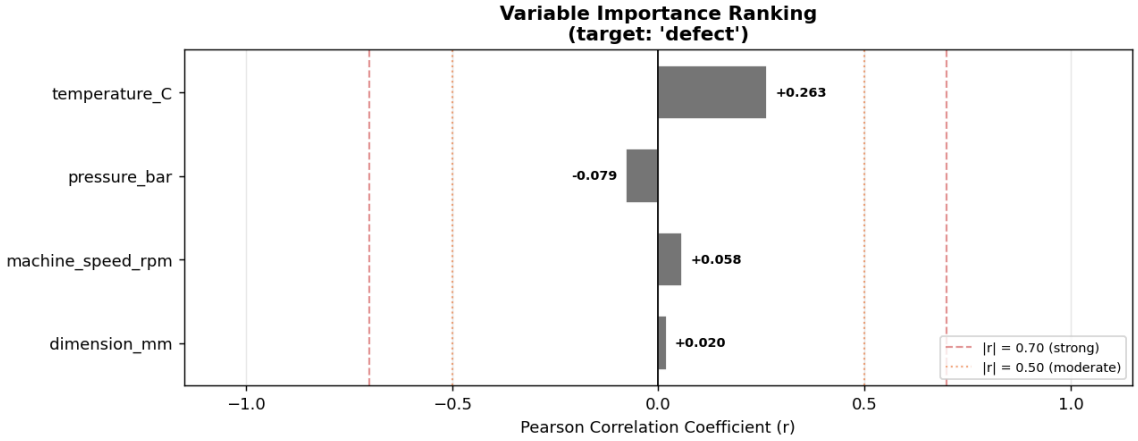


Figure 7: Variable Importance

# 4 Recommendations

Based on the automated analysis, the following actions are recommended:

1. **Address 1 critical finding(s) immediately.** Investigate all points flagged by Western Electric rules outside $3\sigma$ limits. Document findings and corrective actions.
2. **Review 0 warning(s)** with the process engineering team. Schedule corrective actions within the next sprint cycle.
3. **Re-run SigmaFlow** after implementing improvements to verify effectiveness of corrective actions.
4. **Expand data collection** on variables identified in root cause analysis to confirm or refute correlation findings.
5. **Set up real-time monitoring** for datasets with out-of-control processes.

*All findings should be reviewed by a qualified process engineer before implementing corrective actions. Correlation does not imply causation.*

# 5 Conclusion

The SigmaFlow v8 automated analysis successfully processed **1 dataset(s)** and produced the following summary:

| Finding Category | Count |
|---|---|
| Critical (immediate action) | 1 |
| Warning (monitor / address) | 0 |
| Informational (stable) | 1 |

This report was generated automatically by **SigmaFlow v8**. All outputs — figures, insights.json, report.tex, and report.pdf — are available in the `output/` directory. Statistical findings and recommendations should be reviewed by a qualified Six Sigma practitioner or process engineer.